Pathological pallidal beta activity in Parkinson's disease is sustained during sleep and associated with sleep disturbance

Paper: Yin et al. (2023). Human overnight recordings; offline controller simulation.

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Page references below use PDF page numbers, including front matter.

To which general question does this paper relate?

Measurement and sleep-related control objectives. Tests whether a waking motor biomarker can be interpreted the same way during sleep.

How does it answer the general questions?

Measured. Pallidal LFPs were recorded alongside polysomnography in patients off dopaminergic medication, with a dystonia comparison group. Parkinsonian beta remained elevated during wakefulness, NREM and REM sleep and was associated with disturbed sleep (PDF pp. 1–7).

Usable insight. Offline adaptive-DBS simulations using waking beta thresholds produced sleep-dependent stimulation patterns. A controller calibrated during wakefulness can therefore misrepresent therapeutic demand during sleep. These simulations do not demonstrate improved or worsened sleep under an actual adaptive intervention (PDF pp. 7–9).

Model relevance. Supports including operating context and evaluating sleep outcomes separately from motor scores. The results are correlations; suppressing nocturnal beta is not established as a sufficient treatment. No delayed neural model is identified, and the target is pallidal: its biomarker behaviour should not automatically be assigned to STN recordings.

What papers does it cite that could provide more information?

- Zahed et al. (2021), *The neurophysiology of sleep in Parkinson's disease*: sleep mechanisms (ref. 1, PDF p. 10).
- Yin et al. (2023), *Pallidal activities during sleep and sleep decoding in dystonia, Huntington's, and Parkinson's disease*: decoding context (ref. 25, PDF p. 11).